Payal Singh
Senior AI Systems & Reliability Engineer | AI Technical Integrator
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SUMMARY

Senior systems and reliability engineer with 10+ years of experience designing, integrating, and operating mission-critical data and AI-enabled systems. Specializes in integrating AI components into existing cloud and data platforms with explicit guardrails, evaluation, and failure containment. Strong background in PostgreSQL, AWS, applied retrieval-augmented generation (RAG), and AI system integration in regulated and high-reliability environments.

AI SYSTEMS & INTEGRATION PROJECTS

- AI Codebase Conversion & Migration Assessment System: Designed and built an AI-assisted system for assessing and partially automating database migration efforts across large application codebases (Oracle/MySQL/Informix to PostgreSQL). In active use for migration assessments, generating structured risk reports, file-level conversion results, and human-review handoff artifacts.
- Retrieval-Augmented Generation (RAG) Systems: Built RAG systems using PostgreSQL + pgvector and PageIndex, with explicit evaluation, memory scoping, and hallucination mitigation.
- AI Memory Control Plane & Agent Tooling: Designed memory-scoped agents with explicit skills, lifecycle controls, and bounded context to ensure predictable behavior.
- Portfolio & Investment Analysis Agent: Built an AI-assisted portfolio analysis system using retrieval, structured data sources, and deterministic reporting.

PROFESSIONAL EXPERIENCE

Senior Database Reliability Engineer — OmniTI → Credativ → Instaclustr (NetApp) | 2013–Present

- Led integration and operation of large-scale PostgreSQL-backed systems for enterprise and regulated clients, maintaining up to 99.99% uptime.
- Designed and executed complex data platform migrations and modernization initiatives with strict correctness and availability requirements.
- Integrated AI-assisted tooling into database and data pipelines, focusing on reliability, evaluation, and explicit failure reporting rather than autonomous execution.
- Built and operated cloud-native architectures on AWS including Aurora PostgreSQL, S3, IAM, networking, and secure data flows.
- Designed HA/DR strategies and automation that reduced recovery time objectives by up to 70% and improved operational resilience.

TECHNICAL SKILLS

AI Systems Integration, RAG, LLM Orchestration, Evaluation & Guardrails; AWS (Aurora PostgreSQL, S3, IAM, Networking); PostgreSQL (expert), MySQL; Python, SQL, Bash; Cloud & DevOps: Docker, Kubernetes, Terraform, Ansible; Vector Databases: pgvector; Observability, HA/DR, Reliability Engineering

EDUCATION & CERTIFICATIONS

M.S. Computer Science, University of Maryland Baltimore County
B.E. Computer Science, Panjab University
AWS Certified Solutions Architect

WRITING & SPEAKING

Technical writing on AI systems, reliability, and databases: https://medium.com/@reliable-by-design/
Conference speaker on AI-assisted database migrations and reliability-focused AI system design.